

Subarray Sum Queries II

Time limit: 1.00 s

Memory limit: 512 MB

You are given an array of n integers and q queries. In each query, your task is to calculate the maximum subarray sum in the range $[a, b]$.

Empty subarrays (with sum 0) are allowed.

Input

The first line contains two integers n and q : the number of elements and the number of queries.

Then there are n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Finally there are q lines that describe the queries. Each line has two integers a and b .

Output

Print the answer for each query.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $-10^9 \leq x_i \leq 10^9$
- $1 \leq a \leq b \leq n$

Example

Input	Output
8 4 2 5 1 -2 3 -1 -7 1 2 4 2 5 6 7 4 8	6 7 0 3